

Blair Subbaraman

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Education

- 2020-Present PhD, Human Centered Design & Engineering, *University of Washington*, Seattle, WA
Anticipated Graduation: June 2026
Advisor: Nadya Peek
Committee: Mako Hill, Daniela Rosner, Casey Reas
- 2022 MS, Human Centered Design & Engineering, *University of Washington*, Seattle, WA
- 2018 BA, Physics, *Pomona College*, Claremont, CA
Advisor: Dwight Whitaker
Minor: Mathematics

Publications

Refereed Conference and Journal Articles

- 2025 **Blair Subbaraman**, Nathaneal Bursch, and Nadya Peek. It's Not the Shape, It's the Settings: Tools for Exploring, Documenting, and Sharing Physical Fabrication Parameters in 3D Printing. In Proceedings of the 2025 ACM CHI Conference on Human Factors in Computing Systems (CHI '25) 🏆 Best Paper Honorable Mention (Top 5% of all accepted papers)
- 2024 **Blair Subbaraman**, Orlando de Lange, Sam Ferguson, and Nadya Peek. The Duckbot: A System for Automated Imaging and Manipulation of Duckweed. In Plos one 19, no. 1 (2024)
- 2023 **Blair Subbaraman** and Nadya Peek. 3D Printers Don't Fix Themselves: How Maintenance is Part of Digital Fabrication. In Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS '23).
- 2023 **Blair Subbaraman**, Shenna Shim, and Nadya Peek. Forking a Sketch: How the OpenProcessing Community Uses Remixing to Collect, Annotate, Tune, and Extend Creative Code. In Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS '23).
- 2022 **Blair Subbaraman** and Nadya Peek. P5.fab: Direct Control of Digital Fabrication Machines from a Creative Coding Environment. In Proceedings of the 2022 ACM Designing Interactive Systems Conference (DIS '22).

- 2022 Xiaowen Sun, Daniel Gilman Calderón, **Blair Subbaraman**, and Jeffrey A. Burke. An Audience Data-Driven Alternate Reality Storytelling Design. In International Conference on Human-Computer Interaction (C&C '22).

Conference Posters and Juried Demonstrations

- 2024 **Blair Subbaraman** and Nadya Peek. Playing the Print: MIDI-based Fabrication Interfaces to Explore and Document Material Behavior. In Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24).
- 2022 **Blair Subbaraman** and Nadya Peek. Demonstrating p5.fab: Direct Control of Digital Fabrication Machines From a Creative Coding Environment. In Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct).
- 2021 **Blair Subbaraman** and Nadya Peek. Demonstrating p5.fab: Direct Control of Digital Fabrication Machines From a Creative Coding Environment. In ACM Symposium on Computational Fabrication (SCF '21).
- 2021 Gabrielle Benabdallah and **Blair Subbaraman**. BodyTeller: Explorations in Narrative Biosensing, In Society for Social Studies of Science (4S '21).

Other Writing

- 2025 **Blair Subbaraman**. Printed Patterns. Algorithmic Pattern Conference (Alpaca 2025). Sheffield, UK. 2022.
- 2025 Brenden Pelkie et al. (including **Blair Subbaraman**). Democratizing Self-driving Labs Through User-developed Automation Infrastructure. ChemRxiv. 2025.
- 2022 Gabrielle Benabdallah and **Blair Subbaraman**. Explorations in narrative biosensing. Interactions. ACM New York, NY, USA, 2022.

Professional Activities

Talks

** Indicates invited talk*

- 2025 Printed Patterns, *Algorithmic Pattern Conference*, Sheffield, England.

- 2024 Sketching with Machines, *Hackaday Superconference*, Pasadena, CA, USA.
- 2024* Science Jubilee: Open-source Hardware for Laboratory Automation, *Duckweeds, Microbes, and Symbiosis*, University of British Columbia, Vancouver, Canada.
- 2023* Computational Fabrication in Performance Contexts, *UCLA School of Theater, Film, & Television*, Los Angeles, CA, USA.
- 2023 Bridging Creative Code and Digital Fabrication, *Open Source Arts Contributors Conference*, Clinic for Open Source Arts (COSA), University of Denver, CO, USA.
- 2023 Custom Workflow Automation with Jubilee, *Global Community Biosummit 6.0*, Virtual & MIT
- 2022* Niche Experiments with Jubilee: Duckweed Growth Assays as a Case Study, *Duckweeds and Microbes: A New Frontier in Symbiosis Research*, University of Toronto.

Workshops

- 2024, 2025 Pathways to Open-Source Hardware for Laboratory Automation, *University of Washington*, Seattle, Washington, USA, Co-Organizer.
Co-organized two consecutive NSF-sponsored three-day workshop with 25 participants. The workshops gathered an international group of scientists and engineers using open-source technologies for automating scientific experiments to build and share their approaches.
- 2025 3D Printing with p5.fab, *Creative Coding Fest*, Online
Organized multiple online workshops with over 20 educators and hobbyists to introduce 3D printing techniques using Javascript code.
- 2021 Creative Fabrication Workshop, *University of Washington*
Workshop with professional artists and seasoned makers to explore the possibilities of controlling machines with code.

Awards and Honors

- 2025 CHI Best Paper Honorable Mention
- 2024 Processing Foundation Fellowship Finalist

2018 Richard P. Edmunds Physics Prize for outstanding graduating student, Pomona College

Teaching Experience

- 2025 Instructor of Record, HCDE 439: Physical Computing, *University of Washington*
Undergraduate level introduction to engineering and prototyping interactive systems & environments for human-centered applications. 25 students.
- 2021 Graduate Teaching Assistant, HCDE 439: Physical Computing, *University of Washington*
Sole TA, 25 students.
- 2018 Undergraduate Teaching Assistant, PHYS 128: Electronics with Laboratory, *Pomona College*
Transistors and integrated circuits in a variety of applications including operational amplifiers, basic digital circuits, analog/digital conversion and an introduction to microprocessors. 2 TAs, 30 students.
- 2017 Undergraduate Teaching Assistant, PHYS 41: General Physics with Laboratory, *Pomona College*
Calculus-based introduction to Newtonian mechanics and thermodynamics for non-majors. 2 TAs, 25 students.
- 2015, 2016 Undergraduate Teaching Assistant, PHYS 70: Spacetime, Quanta and Entropy with Laboratory, *Pomona College*
Calculus-based introduction to principles of contemporary physics. Topics include conservation laws, special relativity, quantum physics and thermal physics, all viewed from a 21st century perspective. 2 TAs, 30 students.

Mentorship

* Signifies co-authorship on peer-reviewed article

- 2025- Catherine Zhang, Bachelor's Student, Computer Science, University of
Present Washington.
- 2024- Nathaneal Bursch*, Bachelor's Student, Mechanical Engineering, University
2025 of Washington.
- 2022- Sam Ferguson*, Bachelor's Student, Human Centered Design and Engi-
2024 neering, University of Washington.

2022- Shenna Shim*, Bachelor's Student, Human Centered Design and Engineer-
2023 ing, University of Washington.

Service

Reviewing

** Special Recognition for Outstanding Reviews*

2025 ACM Conference on Human Factors in Computing (CHI)*
ACM Conference on Designing Interactive Systems (DIS)*
ACM Symposium on User Interface Software and Technology (UIST)*

2024 ACM Conference on Human Factors in Computing (CHI)*
ACM Conference on Designing Interactive Systems (DIS)*
Halfway to the Future Symposium (HttF)

2023 ACM Conference on Human Factors in Computing (CHI)*
ACM Conference on Designing Interactive Systems (DIS)*

Conference Service

2024 Student Volunteer, ACM Conference on Human Factors in Computing (CHI)

2022 Session Chair, ACM Symposium on Computational Fabrication (SCF)
Student Volunteer, Eyeo Festival

2017 Student Volunteer, Eyeo Festival

Research Positions

2020- Machine Agency
Present with Nadya Peek, University of Washington
Topics: Digital and computational fabrication, laboratory automation

2022 Slip Rabbit Studio
with Timea Tihanyi & Audrey Desjardins, University of Washington
Topics: Data physicalization, ceramic 3D printing

2021 Tactile and Tactical Design Lab
with Daniela Rosner, University of Washington
Topics: Biosensing, wearable technologies

- 2018- Center for Research in Engineering, Media, & Performance (REMAP)
- 2020 with Jeff Burke, UCLA
- Topics: AR theater performance, open source software for the arts

- 2014- Fluids Dynamics Lab
- 2018 with Dwight Whitaker, Pomona College
- Topics: Biological hydrodynamics, scientific instrumentation

Performances & Exhibitions

- 2020 The Invention of Morel, Virtual Performance
- Developer for a workshop and performance exploring distributed remote virtual production during COVID-19 through an adaptation of The Invention of Morel by Adolfo Bioy Casares.

- 2019- The Man in the High Castle, UCLA Freud Playhouse & Virtual
- 2020 AR app development and HCI research for an experimental, immersive theater piece based on the world of the streaming series The Man in the High Castle.

- 2019 @LAs, UCLA Freud Playhouse
- Developer and designer working with directors, cinematographers, and set designers to realize an AI/ML driven pop-up exhibition prototype as part of UCLA's Future Storytelling Summer Institute 2019.

- 2018 The Buzz, UCLA Campus
- Created and deployed the sound platform for a Raspberry Pi based audio/furniture installation in collaboration with UCLA cityLAB.

- 2018 Entropy Bound, UCLA Little Theater
- Developed software for custom wearables to support an experimental play weaving dramaturgy with code.

Selected Synergistic Activities

- 2021- p5.fab
- Present Developed and maintain p5.js library for authoring physical artifacts.

- 2023- Science Jubilee
- Present Developed and maintain open-source software and hardware for laboratory automation.

- 2023- p5.js Math Steward
- 2025 One of ten stewards for issues, contributions, and community discussion related to the Math features of the open source creative coding library p5.js.

- 2018- OpenPTrack Community Manager
- 2021 Managed issues, questions, and discussions for OpenPTrack, an open source person tracking software for the arts.

- 2016- Community Science Projects
- 2018 Led partnership between Pomona College and Fremont Academy of Engineering & Design, a public high school in Pomona, CA.